Flexible and inflexible models in machine learning

I came across a simple question on comparing flexible models (i.e. splines) vs. inflexible models (e.g. linear regression) under different scenarios. The question is:

In general, do we expect the performance of a flexible statistical learning method to perform better or worse than an inflexible method when:

1. The number of predictors p is extremely large, and the number of observations n is small?
2. The variance of the error terms, i.e. $\sigma^2 = \mathrm{Var}(e)$, is extremely high?

I think for (1), when n is small, inflexible models are better (not sure). For (2), I don't know which model is (relatively) better.

machine-learning model

edited Sep 4 '13 at 20:39
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asked Sep 4 '13 at 20:24
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185 9

Generalization error is far from trivial. Unfortunately rules of thumb don't help much in this regard. – Marc Claesen Sep 4 '13 at 20:28

2 Answers

In these 2 situations, comparative performance flexible vs. inflexible model also depends on:

- is true relation y=f(x) close to linear or very non-linear;
- do you tune/constrain flexibility degree of the "flexible" model when fitting it.

If relation is close to linear and you don't constrain flexibility, then linear model should give better test error in both cases because flexible model likely to overfit in both cases.

You can look at it as that:

- In both cases data doesn't contain enough information about true relation (in first case relation is high dimensional and you have not enough data, in second case it corrupted by noise) but
  - linear model brings some external prior information about true relation (constrain class of fitted relations to linear ones) and
  - that prior info turns out to be right (true relation is close to linear).
- While flexible model doesn't contain prior information (it can fit anything), so it fits to noise.

If however true relation is very non-linear, it's hard to say who will win (both will loose :)).

If you tune/constrain degree of flexibility and do it in a right way (say by cross-validation), then flexible model should win in all cases.

edited Nov 28 '13 at 2:20

answered Nov 28 '13 at 2:02
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391 1 9

Well, for the second part, I think more more flexible model will try to fit the model hard and training data contains a high noise, so flexible model will also try to learn that noise and will result in more test error. I know the source of this question as I'm also reading the same book :)

answered Oct 28 '13 at 16:08
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80 4